ClinGen's Variant Curation Interface (VCI) imports evidence data in standard formats via the Linked Data Hub (LDH)

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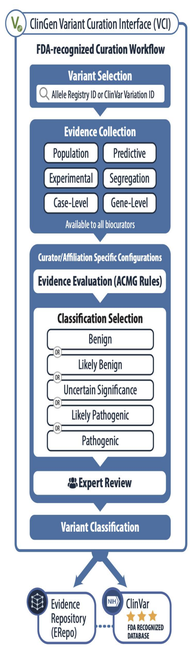
Abstract

The NIH-funded Clinical Genome Resource Consortium (ClinGen) has developed a suite of tools that support variant classification. Several of these tools focus on the application of evidence criteria and classification of variants based on the ACMG/AMP sequence variant classification guidelines, and then the dissemination of those classifications to the global community. Among these tools is the Variant Curation Interface (VCI: https://curation.clinicalgenome.org), an open-source variant classification platform that supports the FDA-recognized human variant curation process by ClinGen Variant Curation Expert Panels (VCEPs). The Linked Data Hub (LDH: https://ldh.genome.network/srvc) also supports variant curation, by providing efficient access to collated and standardized variant information from different data sources, data provenance information, and links to data sources made available through RESTful APIs. The LDH ingests evidence from key resources in non-standard formats and provides the VCI with this evidence in a structured, machine-readable standardized format to facilitate evidence data incorporation into variant curation pipelines.

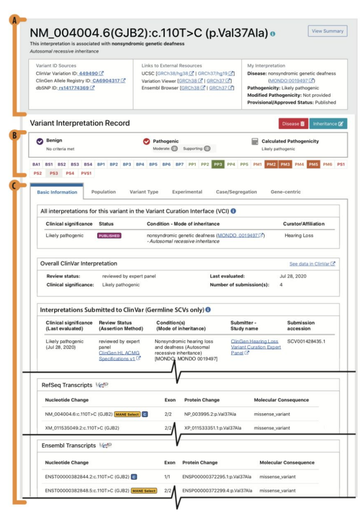
One challenge for variant curation platforms such as the VCI involves identifying these critically valuable resources, accurately presenting the relevant evidence for specific variants, and providing evidence relevant to a given disease in a structured format for variant curators to review and then use in variant classification. Here we present how the LDH and VCI have addressed this challenge. Specifically we describe two cases where the LDH acquired data from external resources (in this case the pathogenicity predictor REVEL, and the Genome Aggregation Database (gnomAD) version 4.1 release), and provided it such that the VCI can appropriately access and display the data to curators within the curation interface. We describe how these use cases highlight the need for data standardization, version control, and user-driven feedback.

Variant Curation Evidence and Curation Workflow in the VCI



Left: Overview of the ClinGen Variant Curation Process using the VCI. Biocurators select a variant and evaluate evidence. VCI viewers may view all evidence available for any variant using the VCI. ClinGen expert panels disseminate their final classifications through two community resources: ClinVar and the ClinGen Evidence Repository (ERepo).

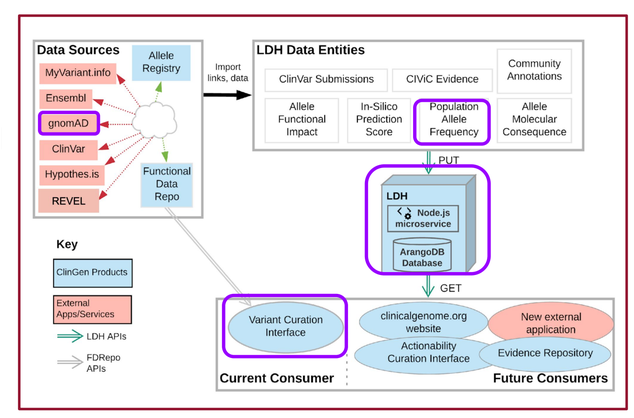
Right: The VCI "Basic Information" view. The VCI has six tab views that collate and display variant information from external and internal sources to biocurators. (A) The top title view is always viewable and shows the variant title, and key curation information for the record. (B) The criteria bar displays evaluated criteria and the calculated pathogenicity. (C) The basic information tab displays data from ClinVar, RefSeq, and Ensembl.



The Linked Data Hub (LDH)

Implementation of a Linked Data Hub strategy, as shown to the right, provides for aggregation of structured data from multiple sources to serve a multitude of utilities:

- Adapts W3C Web Annotation Data Model recommendations
- Stores key select data, excerpts or full data
- Each entity like gnomAD and REVEL will have its own data-model
- LDH Core and the supporting microservices such as LDH Data Storage Service, LDH-Alias service will enable efficient ingestion, storing and retrieval of the linked data
- The LDH permanent links via APIs allow programmatic integration to the VCI



Above: The LDH ecosystem. The key data pipeline further discussed in this poster are outlined in purple.

Example display of REVEL predictor scores

The initial implementation of the LDH/VCI integration has been to display REVEL meta-predictor scores associated with the variant, the associated transcript for the score, and the MANE status of the transcript. Currently this information is not available via API call from any source. Additionally, anyone can query the LDH using a unique variant identifier ID to access all of this key information

ClinGen Predictors

Source	Score Range	Score	Transcript	Impact Threshold	Prediction
REVEL (meta-predictor)	0 to 1	0.982	ENST00000268124 MANE Select	>0.75	higher score = higher pathogenicity
		0.982	ENST00000442287		
			Hide non-MANE transcript scores		

Example of REVEL data in the VCI sourced from the LDH

Variant: NM_002693.3(POLG):c.2794C>T (p.His932Tyr) / ClinVar Variation ID: 13500 / ClinGen Allele Registry ID: CA256891

Example display of gnomAD v4.1 population data

The initial implementation of the LDH/VCI integration will involve displaying gnomAD v4.1 population allele frequency data in the VCI. Further improvements will involve additional display of gnomAD v4.1 data such as the GroupMax filtering allele frequency, gene constraint scores, and hemizygous counts for X-linked genes. All of this gnomAD v4.1 information will be available via the LDH to any user who queries using a unique identifier ID.

gnomAD v4 GroupMax Filtering Allele Frequency

	Exomes	Genomes	Total
Filters	Pass	Pass	
Allele Count	425	37	462
Allele Number	1461894	152286	1614180
Allele Frequency	0.0002907187525224127	0.00024296389687824226	0.0002862134334460841
Grpmax Filtering AF (95% confidence)	0.00033554	0.0003295100000000001	0.00034158
Grpmax Filtering AF Group	Non-Finnish European	Non-Finnish European	Non-Finnish European
Number of homozygotes	0	0	0

ClinGen guidance for using gnomAD v4 data

gnomAD 17-42911391-C-T (GRCh38) Version: 4.1 See data in gnomAD

Exomes Filter: Pass

Genomes Filter: Pass

Genetic Ancestry Group	Allele Count	Allele Number	Number of Homozygotes	Allele Frequency
Non-Finnish European	437	1180030	0	0.00037032956789234174
Remaining individuals	10	62512	0	0.00015996928589710776
Admixed American	9	60016	0	0.00014996001066382297
African/African-American	6	75032	0	0.00007996588122401109
Amish	0	912	0	0
South Asian	0	91080	0	0
Middle Eastern	0	6062	0	0
Finnish	0	64038	0	0
East Asian	0	44890	0	0
Ashkenazi Jewish	0	29608	0	0
XX	**264**	**812470**	**0**	**0.00032493507452582864**
XY	**198**	**801710**	**0**	**0.0002469720971423582**
Total	**462**	**1614180**	**0**	**0.0002862134334460841**

Example of gnomAD v4.1 data in the VCI sourced from the LDH

Variant: NM_000151.4(G6PC1):c.1039C>T (p.Gln347Ter) / ClinVar Variation ID:12000 / CA ID: CA256179

Supporting Data Standards for Evidence

The ClinGen variant curation suite of tools (including the VCI and LDH) supports data standardization for variant curation evidence in several different ways:

- By using and supporting globally consistent and unique variant identification systems to associate evidence with variants
- By cross-referencing evidence with these unique identifiers whenever possible to ensure broad access to evidence for curators
- By following Findable, Accessible, Interoperable, and Reusable (FAIR) Principles
- By working with resources such as the Global Alliance for Genomics & Health (GA4GH) and the Monarch Initiative's Scientific Evidence and Provenance Information Ontology (SEPIO), to create and use data standards and data models

Acknowledgements

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